

Lab Report Assignment

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CISP5302

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Abstract

The Portuguese bank aims to enhance the effectiveness of term deposit promotion campaigns through direct marketing. However, pinpointing key factors influencing customer subscription via phone calls poses a challenge. Utilizing SAS Enterprise Miner, the BANK dataset analysis employs predictive modeling and segmentation to uncover subscription drivers. With 17 variables and 4,521 observations, 'y' indicates term deposit subscription. SEMMA framework guides the exploration, modification, modeling, and assessment stages, ensuring robust analysis. This methodology offers insights into customer behavior, empowering the bank to optimize marketing strategies and elevate term deposit subscription rates. Further research in predictive modeling could explore emerging trends and innovative techniques to enhance campaign efficacy and customer engagement.

Business Problem

The Portuguese bank wants to increase the term deposit promotion campaigns' direct marketing efficacy. Nevertheless, it has trouble pinpointing the crucial elements that affect customers' decision to sign up for term deposits via phone call marketing.

Using SAS Enterprise Miner's data mining features, we recommend studying the BANK dataset to obtain insight on the factors impacting client subscription. By using data analysis, segmentation, and predictive modeling to the dataset, we can identify patterns and associations that will optimize the efficacy of future marketing initiatives.

The dataset includes 17 variables (client demographics, financial data, campaign specifics, and outcomes) and 4,521 observations. Whether the client subscribed to the term deposit ('yes') or not ('no') is indicated by the target variable 'y'. We will investigate the effects of important input variables on subscription likelihood, including age, work type, marital status, education level, contact communication type, and past campaign outcomes.

Methodology - SEMMA framework

The analysis of the BANK dataset for term deposit subscription prediction starts with sampling, in which a representative subset is selected, using the SEMMA framework. During the exploring phase, exploratory data analysis finds possible elements that could affect subscription. In the Modify step, missing values and outliers are addressed along with preprocessing of the data. Several methods, including decision trees and regression, are used in the modeling phase to create predictive models. Lastly, model performance is assessed in Assessing to identify the best course of action. The Portuguese banking institution may optimize direct marketing campaigns and increase term deposit subscription rates by using this methodology, which provides insights into customer behavior.

Data Exploration

Model Roles :-

'y' denotes a term deposit subscription, and inputs include age, marital status, education level, work type, contact communication method, and previous campaign results. "Poutcome" is rejected since it refers to a previous campaign's outcome and might not be relevant to the current.

Name	Role	Level	Report	Order	Drop	Lower Limit	Upper Limit
age	Input	Interval	No		No	.	.
balance	Input	Interval	No		No	.	.
campaign	Input	Interval	No		No	.	.
contact	Input	Binary	No		No	.	.
day	Input	Interval	No		No	.	.
default	Input	Binary	No		No	.	.
duration	Input	Interval	No		No	.	.
education	Input	Nominal	No		No	.	.
housing	Input	Binary	No		No	.	.
job	Input	Nominal	No		No	.	.
loan	Input	Binary	No		No	.	.
marital	Input	Nominal	No		No	.	.
month	Input	Nominal	No		No	.	.
pdays	Input	Interval	No		No	.	.
poutcome	Rejected	Nominal	No		No	.	.
previous	Input	Interval	No		No	.	.
y	Target	Nominal	No		No	.	.

Figure 1. Variables in Bank Dataset

Data Type/Measurement Levels :-

The dataset includes numerical variables (campaign, balance) as well as categorical ones (day of week, education level). Measurement levels vary from nominal (e.g., marital status, job type) to interval/ratio (e.g., age, balance).

Balanced/Imbalanced dataset :-

A potential imbalance in the dataset may affect the effectiveness of the model, given the majority of respondents have their term deposit subscription categorized as "no".

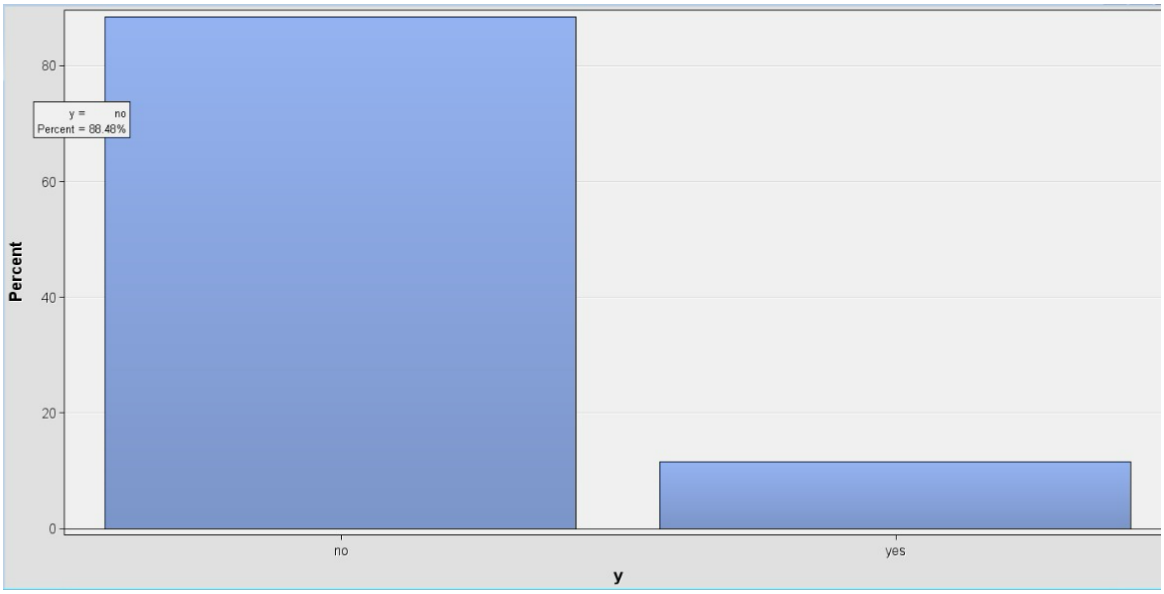


Figure 2. Distribution of Target Status

Missing data :-

There are noticeable cases of missing values for variables including education, contact, and outcome. These variables must be handled carefully in order to avoid bias in analysis.

Name	Role	Level	Report	Order	Drop	Lower Limit	Upper Limit	Number of Levels	Percent Missing	Minimum	Maximum	Mean	Standard Deviation	Skewness
age	Input	Interval	No		No	.	.	.	0	19	87	41.1701	10.57621	0.699501
balance	Input	Interval	No		No	.	.	.	0	-3313	71188	1422.658	3009.638	6.596431
campaign	Input	Interval	No		No	.	.	.	0	1	50	2.79363	3.109807	4.743914
contact	Input	Binary	No		No	.	.	2	29.28556	.	.	.	.	.
day	Input	Interval	No		No	.	.	.	0	1	31	15.91528	8.247667	0.094627
default	Input	Binary	No		No	.	.	2	0	.	.	.	.	.
duration	Input	Interval	No		No	.	.	.	0	4	3025	263.9613	259.8566	2.77242
education	Input	Nominal	No		No	.	.	3	4.136253	.	.	.	.	.
housing	Input	Binary	No		No	.	.	2	0	.	.	.	.	.
job	Input	Nominal	No		No	.	.	11	0.840522	.	.	.	.	.
loan	Input	Binary	No		No	.	.	2	0	.	.	.	.	.
marital	Input	Nominal	No		No	.	.	3	0	.	.	.	.	.
month	Input	Nominal	No		No	.	.	12	0	.	.	.	.	.
pdays	Input	Interval	No		No	.	.	.	0	-1	871	39.76664	100.1211	2.717071
poutcome	Rejected	Nominal	No		No	.	.	2	86.30834	.	.	.	.	.
previous	Input	Interval	No		No	.	.	.	0	0	25	0.542579	1.693562	5.875259
y	Target	Nominal	No		No	.	.	2	0	.	.	.	.	.

Figure 3. Missing values of all variables

Variance and standard deviation :-

A high standard deviation is observed in the 'balance' variable, indicating significant variation in account balances among clients, which is essential for precise model estimation.

Skewness :-

The variable 'balance' exhibits a significant degree of skewness, indicating a deviation from a symmetric distribution. This may necessitate a transformation in order to conduct reliable analysis.

Outliers :-

Outliers draw attention to notable variations in client behavior , in the case of the "no" subscription class in particular, call for cautious handling when modeling.

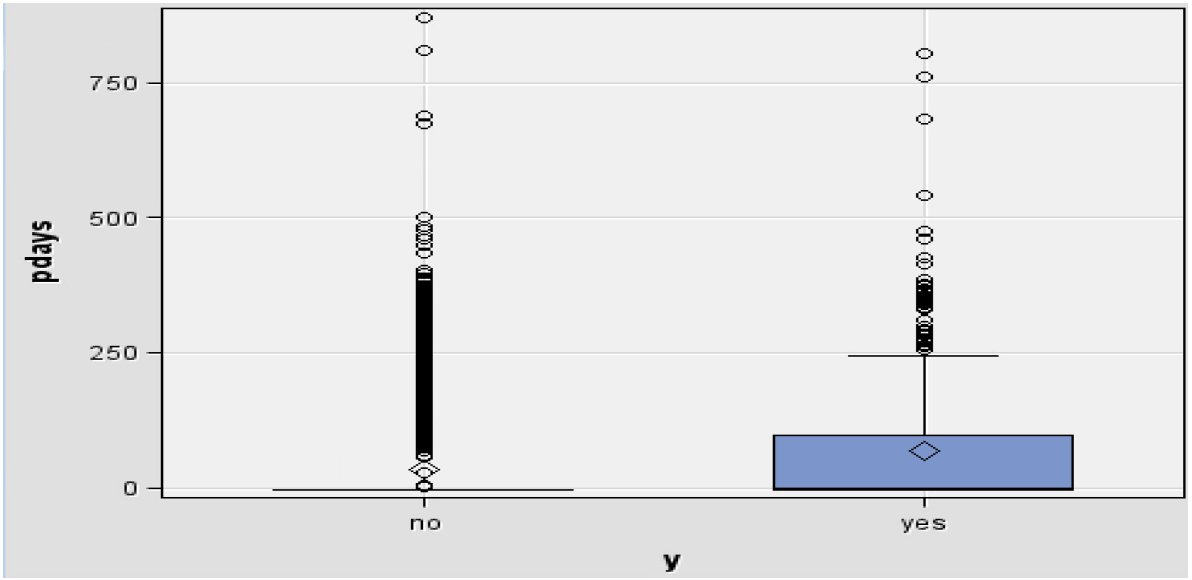


Figure 4. Box plot of pdays vs y

Multicollinearity :-

Identifying and resolving multicollinearity between input variables improves the interpretability and generalizability of the model and ensures accurate estimation.

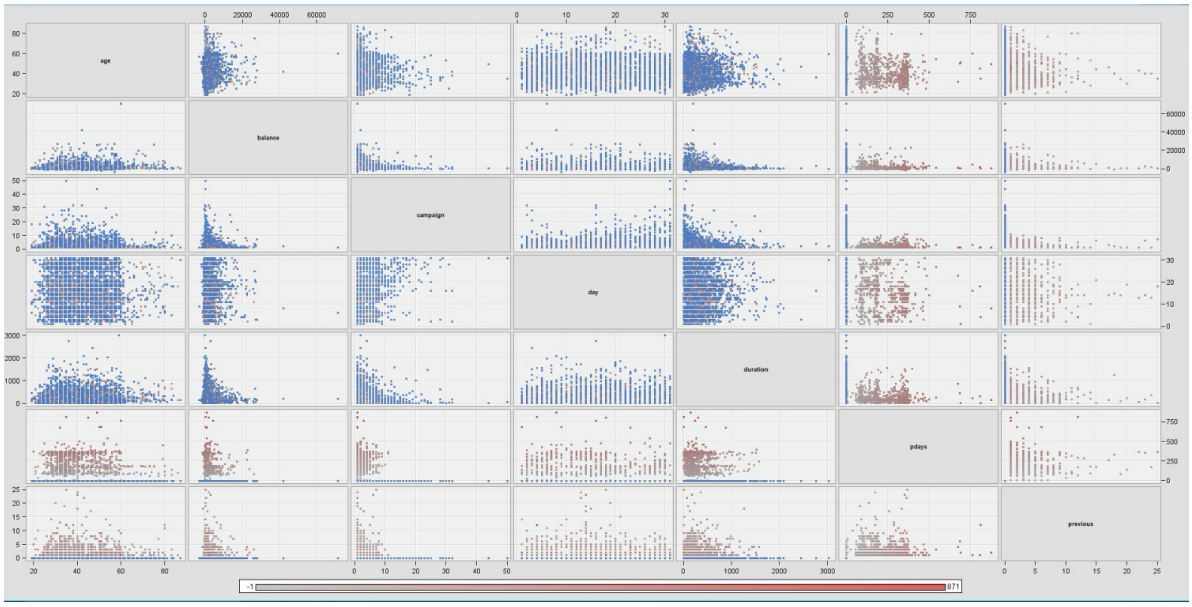


Figure 5. Matrix plot of all input variables group with status

The selected sampling method :-

The model's ability to capture variability and patterns for reliable predictions is improved by stratified sampling, which guarantees representation of various subscription outcomes.

Property	Value
General	
Node ID	Part
Imported Data	...
Exported Data	...
Notes	...
Train	
Variables	...
Output Type	Data
Partitioning Method	Stratified
Random Seed	7642
Data Set Allocations	
Training	40.0
Validation	30.0
Test	30.0
Report	
Interval Targets	Yes
Class Targets	Yes
Status	
Create Time	28/03/2024, 02:09
Run ID	93b5ced8-4543-4ca6-9560-bf573bfd2fcb
Last Error	
Last Status	Complete
Last Run Time	10/04/2024, 17:01
Run Duration	0 Hr. 0 Min. 6.95 Sec.
Grid Host	
User-Added Node	No

Figure 6. Stratified Sampling Method

Data partition creation of model sets

A validation set (30%) helps choose the best model, while a training set (40%) is used to build a set of data mining models. Finally, the Test Set (30%) is used to assess the selected model's efficacy.

Data Modification

Mean imputation was implemented to address missing values, enhancing data quality based on observations from the exploration stage. This method fills in gaps by replacing missing values with the mean of the respective variables, ensuring completeness in the dataset.

Variable Name	Impute Method	Imputed Variable	Impute Value	Role	Measurement Level	Label	Number of Missing for TRAIN
contact	TREE	IMP_contact		INPUT	BINARY	contact	543
education	TREE	IMP_education		INPUT	NOMINAL	education	67
job	TREE	IMP_job		INPUT	NOMINAL	job	12

Figure 7. Imputation Summary

In order to mitigate the dataset's skewness and improve its normal distribution and prediction models accuracy, transformation techniques such logarithmic and square root transformations were applied.

Source	Method	Variable Name	Formula	Number of Levels	Non Missing	Missing	Minimum	Maximum	Mean	Standard Deviation	Skewness	Kurtosis	Label
Input	Original	balance		.	1808	0	-3313	71188	1426.061	3207.382	8.426005	136.7879	balance
Input	Original	campaign		.	1808	0	1	32	2.862279	3.124488	3.95578	22.02707	campaign
Input	Original	day		.	1808	0	1	31	15.90708	8.186887	0.089021	-1.03321	day
Input	Original	duration		.	1808	0	5	2456	268.7074	270.0797	2.581189	9.439653	duration
Input	Original	pdays		.	1808	0	-1	871	38.79038	102.5843	3.072782	11.28663	pdays
Input	Original	previous		.	1808	0	0	20	0.544801	1.699697	5.009734	34.34795	previous
Output	Formula	Tr_balance	LOG10(balance + 1)	.	1670	138	0	4.852413	2.538561	1.021538	-1.12447	0.90616	
Output	Formula	Tr_campaign	LOG10(campaign)	.	1808	0	0	1.50515	0.319498	0.31407	0.853355	0.346045	
Output	Formula	Tr_day	SQRT(day)	.	1808	0	1	5.567764	3.828279	1.118951	-0.38769	-0.76426	
Output	Formula	Tr_duration	LOG10(duration)	.	1808	0	0.69897	3.390228	2.248701	0.418248	-0.44633	0.708935	
Output	Formula	Tr_pdays	LOG(pdays + 1)	.	310	1498	0.693147	6.770789	5.250297	0.777605	-2.74974	13.03342	
Output	Formula	Tr_previous	LOG10(previous + 1)	.	1808	0	0	1.322219	0.093637	0.228605	2.54841	5.901985	

Figure 8. Transformation Statistics

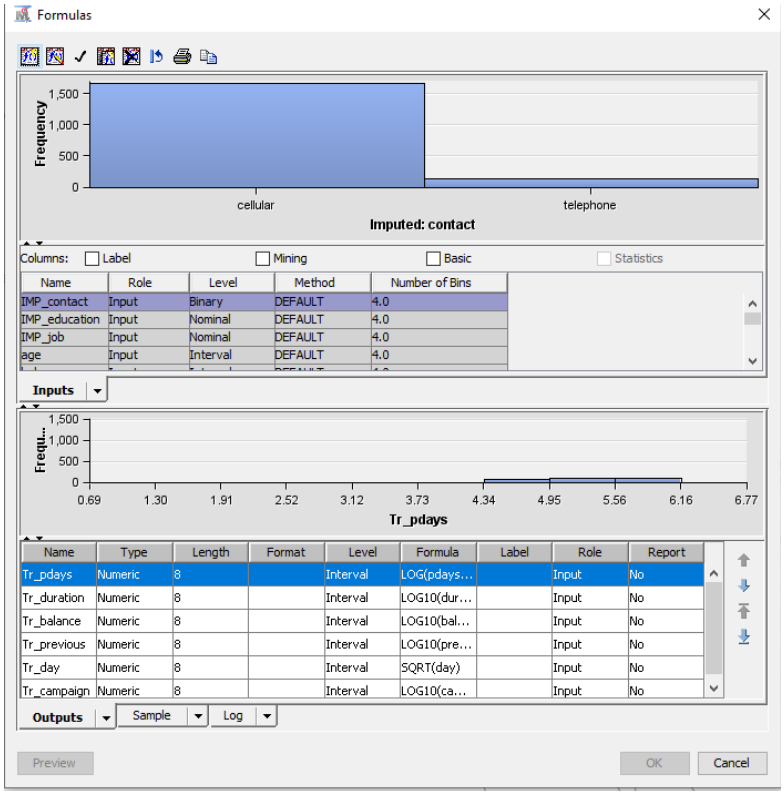


Figure 9. Transformation using Log and SQRT

Data Modelling

Regression

Development of models

Five regression models were developed using various techniques like R2, Chi², imputation, transformations, and backward selection to predict term deposit subscriptions. The "impute with default" method emerged as the most effective predictor based on performance metrics such as misclassification rate, ROC, classification chart, and true positive

Model performance

Table 1: Regression models performance

Model variations	ROC Index	Cu. Lift	Scope %	True – %	False – %	True + %	False + %
Reg-IMP-Default	0.884	3.191192	**20**	**98.4987**	86.4516	**13.5484**	1.5013
Reg-IMP-Tr-Default	0.653	1.664985	**20**	**98.6656**	87.0968	**12.9032**	1.3344
Reg-IMP-Backward	0.877	3.320129	20	98.4987	87.7419	12.2581	1.5013
Reg-IMP-Tr-Backward	0.815	2.514272	20	89.9917	65.1613	34.8387	10.0083
Reg-IMP-Backward-Chi2	0.876	3.191192	20	98.4987	87.0968	12.9032	1.5013
Reg-IMP-Backward-Tr-Chi2	0.882	3.352363	20	98.3319	81.2903	18.7097	1.6681
Reg-IMP-Backward-R2	0.852	2.627092	20	98.8324	88.3871	11.6129	1.1676
Reg-IMP-Backward-Tr-R2	0.653	2.160852	20	99.8332	98.7097	1.2903	0.1668

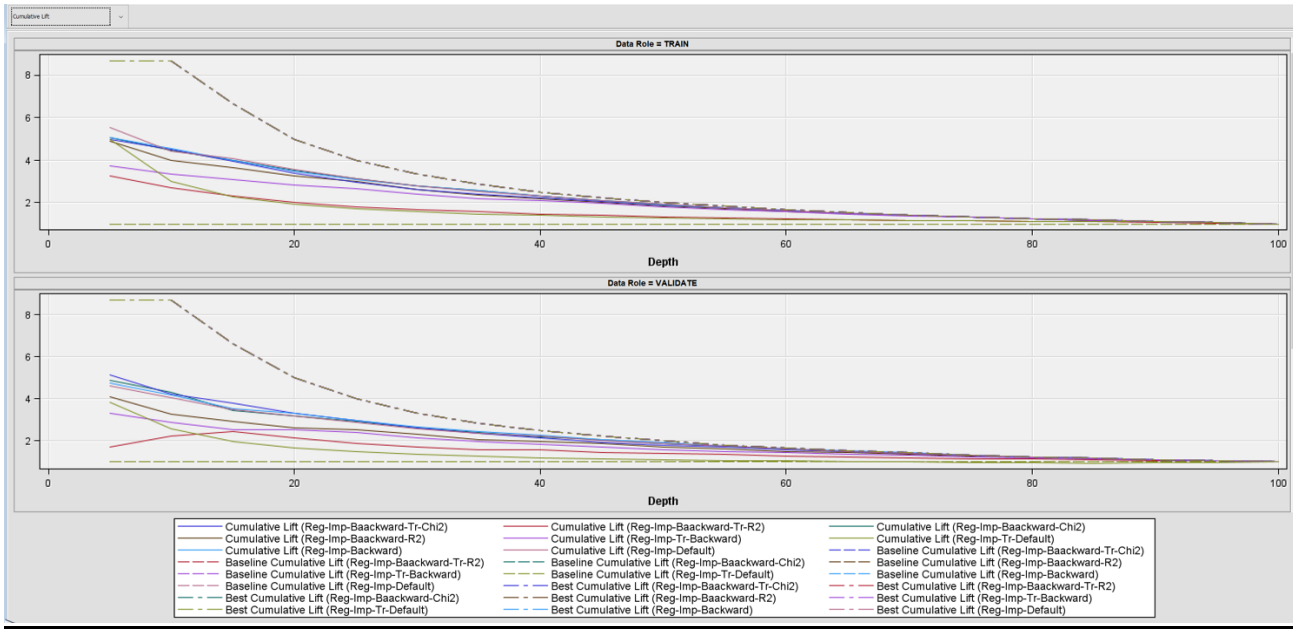


Figure 10. Score Rankings Overlay: y (Cumulative lift)

Event Classification Table
Model Selection based on Train: Roc Index (_AUR_)

Model Node	Model Description	Data Role	Target	Target Label	False Negative	True Negative	False Positive	True Positive
Reg	Reg-Imp-Default	TRAIN	y	y	154	1567	33	54
Reg	Reg-Imp-Default	VALIDATE	y	y	134	1181	18	21
Reg2	Reg-Imp-Backward	TRAIN	y	y	163	1571	29	45
Reg2	Reg-Imp-Backward	VALIDATE	y	y	136	1181	18	19
Reg3	Reg-Imp-Tr-Default	TRAIN	y	y	176	1587	13	32
Reg3	Reg-Imp-Tr-Default	VALIDATE	y	y	135	1183	16	20
Reg4	Reg-Imp-Tr-Backward	TRAIN	y	y	118	1433	167	90
Reg4	Reg-Imp-Tr-Backward	VALIDATE	y	y	101	1079	120	54
Reg5	Reg-Imp-Baackward-R2	TRAIN	y	y	167	1569	31	41
Reg5	Reg-Imp-Baackward-R2	VALIDATE	y	y	137	1185	14	18
Reg6	Reg-Imp-Baackward-Chi2	TRAIN	y	y	164	1565	35	44
Reg6	Reg-Imp-Baackward-Chi2	VALIDATE	y	y	135	1181	18	20
Reg7	Reg-Imp-Baackward-Tr-R2	TRAIN	y	y	207	1598	2	1
Reg7	Reg-Imp-Baackward-Tr-R2	VALIDATE	y	y	153	1197	2	2
Reg8	Reg-Imp-Baackward-Tr-Chi2	TRAIN	y	y	156	1561	39	52
Reg8	Reg-Imp-Baackward-Tr-Chi2	VALIDATE	y	y	121	1178	21	34

Figure 1. Classification table of Model Selection

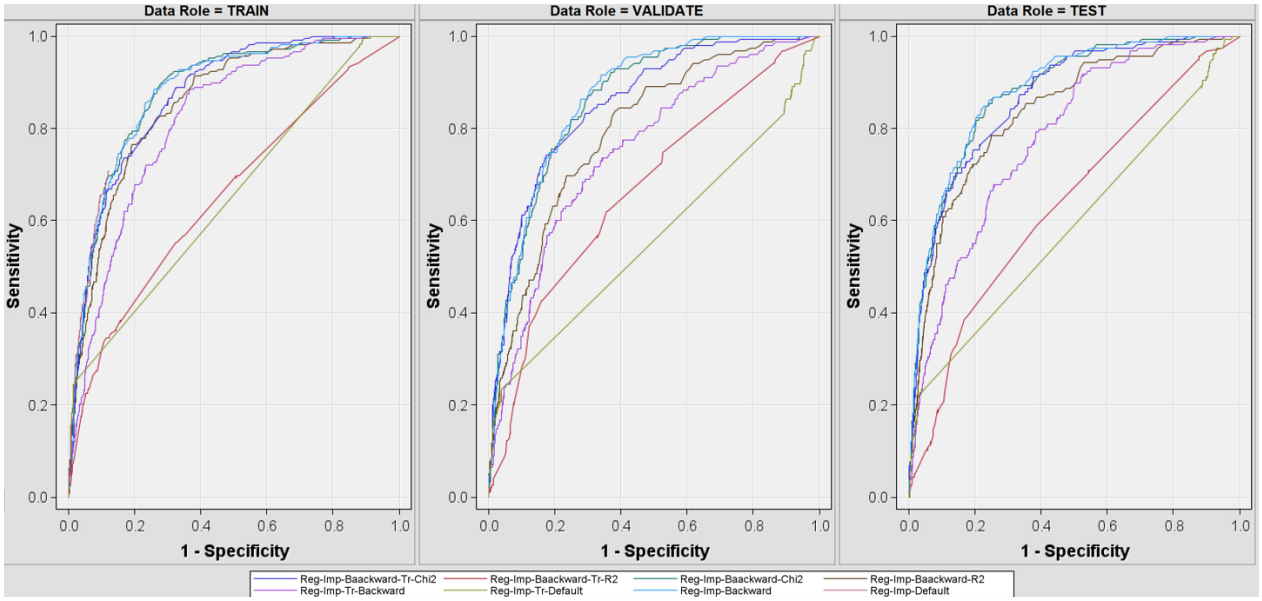


Figure 2. Roc Chart For Regression Model

Fit Statistics
Model Selection based on Train: Roc Index (_AUR_)

Selected Model	Model Node	Model Description	Train: Roc Index	Train: Average Squared Error	Train: Misclassification Rate	Valid: Average Squared Error	Valid: Misclassification Rate
Y	Reg	Reg-Imp-Default	0.884	0.07723	0.10343	0.08385	0.11226
	Reg2	Reg-Imp-Backward	0.877	0.07940	0.10619	0.08235	0.11374
	Reg6	Reg-Imp-Baackward-Chi2	0.876	0.08022	0.11007	0.08212	0.11300
	Reg8	Reg-Imp-Baackward-Tr-Chi2	0.868	0.07857	0.10785	0.07998	0.10487
	Reg5	Reg-Imp-Baackward-R2	0.852	0.08271	0.10951	0.08882	0.11152
	Reg4	Reg-Imp-Tr-Backward	0.815	0.10817	0.15763	0.11634	0.16322
	Reg7	Reg-Imp-Baackward-Tr-R2	0.653	0.09679	0.11560	0.09809	0.11448
	Reg3	Reg-Imp-Tr-Default	0.653	0.09781	0.10454	0.10830	0.11152

Figure 3. Model selection statistics

Chosen Regression equation

The regression method that was chosen was "Impute default," which involved using the Logistic Equation Regression to calculate the likelihood that a consumer will subscribe to a term deposit.

Analysis of Maximum Likelihood Estimates

Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq	Standardized Estimate	Exp(Est)
Intercept		1	-2.5269	0.6381	15.68	<.0001		0.080
IMP_contact	cellular	1	0.0168	0.1729	0.01	0.9228		1.017
IMP_education	primary	1	-0.0445	0.1937	0.05	0.8181		0.956
IMP_education	secondary	1	-0.0473	0.1393	0.12	0.7343		0.954
IMP_job	admin.	1	0.0682	0.2820	0.06	0.8088		1.071
IMP_job	blue-collar	1	-0.1101	0.2438	0.20	0.6515		0.896
IMP_job	entrepreneur	1	0.0899	0.4600	0.04	0.8450		1.094
IMP_job	housemaid	1	0.1117	0.5123	0.05	0.8274		1.118
IMP_job	management	1	0.00148	0.2438	0.00	0.9952		1.001
IMP_job	retired	1	1.2351	0.3584	11.87	0.0006		3.439
IMP_job	self-employed	1	-0.6010	0.4579	1.72	0.1893		0.548
IMP_job	services	1	0.3397	0.3018	1.27	0.2604		1.405
IMP_job	student	1	0.0917	0.4847	0.04	0.8499		1.096
IMP_job	technician	1	-0.3607	0.2609	1.91	0.1667		0.697
age		1	-0.0170	0.0110	2.38	0.1228	-0.0997	0.983
balance		1	-3.64E-6	0.000022	0.03	0.8693	-0.00643	1.000
campaign		1	-0.0121	0.0333	0.13	0.7155	-0.0209	0.988
day		1	0.0193	0.0121	2.53	0.1114	0.0870	1.019
default	no	1	-0.5375	0.2812	3.65	0.0559		0.584
duration		1	0.00393	0.000292	181.88	<.0001	0.5858	1.004
housing	no	1	0.3165	0.1085	8.51	0.0035		1.372
loan	no	1	0.3590	0.1485	5.84	0.0157		1.432
marital	divorced	1	0.0478	0.1837	0.07	0.7948		1.049
marital	married	1	-0.0826	0.1318	0.39	0.5307		0.921
month	apr	1	0.0594	0.2870	0.04	0.8360		1.061
month	aug	1	-0.4932	0.2669	3.42	0.0646		0.611
month	dec	1	0.6933	0.6782	1.05	0.3066		2.000
month	feb	1	0.3235	0.3440	0.88	0.3471		1.382
month	jan	1	-0.8368	0.4776	3.07	0.0797		0.433
month	jul	1	-0.5479	0.2592	4.47	0.0345		0.578
month	jun	1	-0.2854	0.2747	1.08	0.2990		0.752
month	mar	1	1.2890	0.5454	5.59	0.0181		3.629
month	may	1	-0.9124	0.2421	14.20	0.0002		0.402
month	nov	1	-0.7191	0.3073	5.48	0.0193		0.487
month	oct	1	1.0905	0.3962	7.57	0.0059		2.976
pdays		1	0.00277	0.000875	9.98	0.0016	0.1564	1.003
previous		1	0.0254	0.0510	0.25	0.6177	0.0238	1.026

Regarding the most impactful predictors in the model,

Logit p = -2.5269(Intercept)+0.00393(duration)

Decision Tree

Development of models

Multiple versions of decision tree models were developed, including DT_Def, DT_Imp, DT_Imp_Chi2, and DT_Imp_Tr_Chi2, with variations in imputation methods and feature selection techniques. Additionally, models DT_Imp_Tr and DT_Imp_R2 were evaluated based on Cumulative Lift, True/False percentages, and ROC Index.

Performance of models

Table 2: Decision tree models performance//

Model variations	ROC Index	Cu. Lift	Scope %	True – %	False – %	True + %	False + %
DT_Def	0.87	3.174342	**20**	**99.0826**	84.5161	**15.4839**	0.9174
DT_Imp	0.87	3.174342	20	**99.0826**	84.5161	**15.4839**	0.9174
DT_Imp_Chi2	0.87	3.174342	20	**99.0826**	84.5161	**15.4839**	0.9174
DT_Imp_Tr_Chi2	0.87	3.174342	20	**99.0826**	84.5161	**15.4839**	0.9174
DT_Imp_Tr	0.824	3.01114	20	99.2492	88.3871	11.6129	0.7506
DT_Imp_R2	0.5	0	20	100	100	0	0
DT_Imp_Tr_R2	0.5	0	20	100	100	0	0

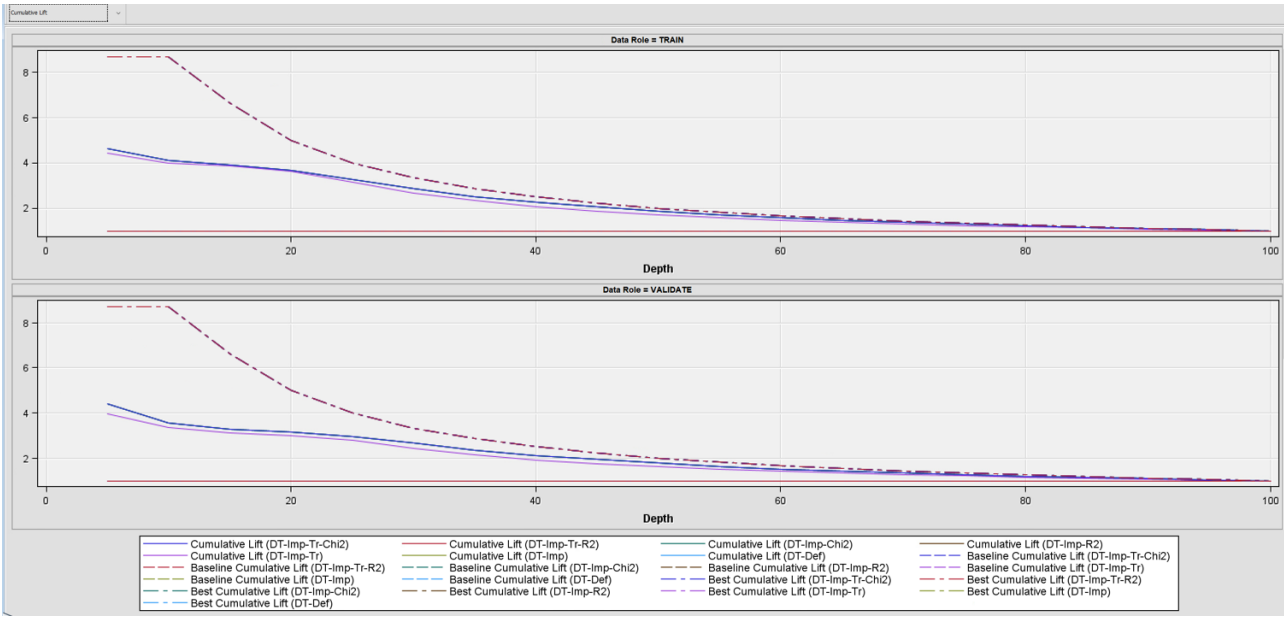


Figure 14. Score Rankings Overlay: y (cumulative lift)

Event Classification Table
Model Selection based on Train: Roc Index (_AUR_)

Model Node	Model Description	Data Role	Target	Target Label	False Negative	True Negative	False Positive	True Positive
Tree	DT-Def	TRAIN	y	y	185	1594	6	23
Tree	DT-Def	VALIDATE	y	y	131	1188	11	24
Tree2	DT-Imp	TRAIN	y	y	185	1594	6	23
Tree2	DT-Imp	VALIDATE	y	y	131	1188	11	24
Tree3	DT-Imp-Tr	TRAIN	y	y	188	1593	7	20
Tree3	DT-Imp-Tr	VALIDATE	y	y	137	1190	9	18
Tree4	DT-Imp-R2	TRAIN	y	y	208	1600	0	0
Tree4	DT-Imp-R2	VALIDATE	y	y	155	1199	0	0
Tree5	DT-Imp-Chi2	TRAIN	y	y	185	1594	6	23
Tree5	DT-Imp-Chi2	VALIDATE	y	y	131	1188	11	24
Tree6	DT-Imp-Tr-R2	TRAIN	y	y	208	1600	0	0
Tree6	DT-Imp-Tr-R2	VALIDATE	y	y	155	1199	0	0
Tree7	DT-Imp-Tr-Chi2	TRAIN	y	y	185	1594	6	23
Tree7	DT-Imp-Tr-Chi2	VALIDATE	y	y	131	1188	11	24

Figure 15. Classification table of Model Selection

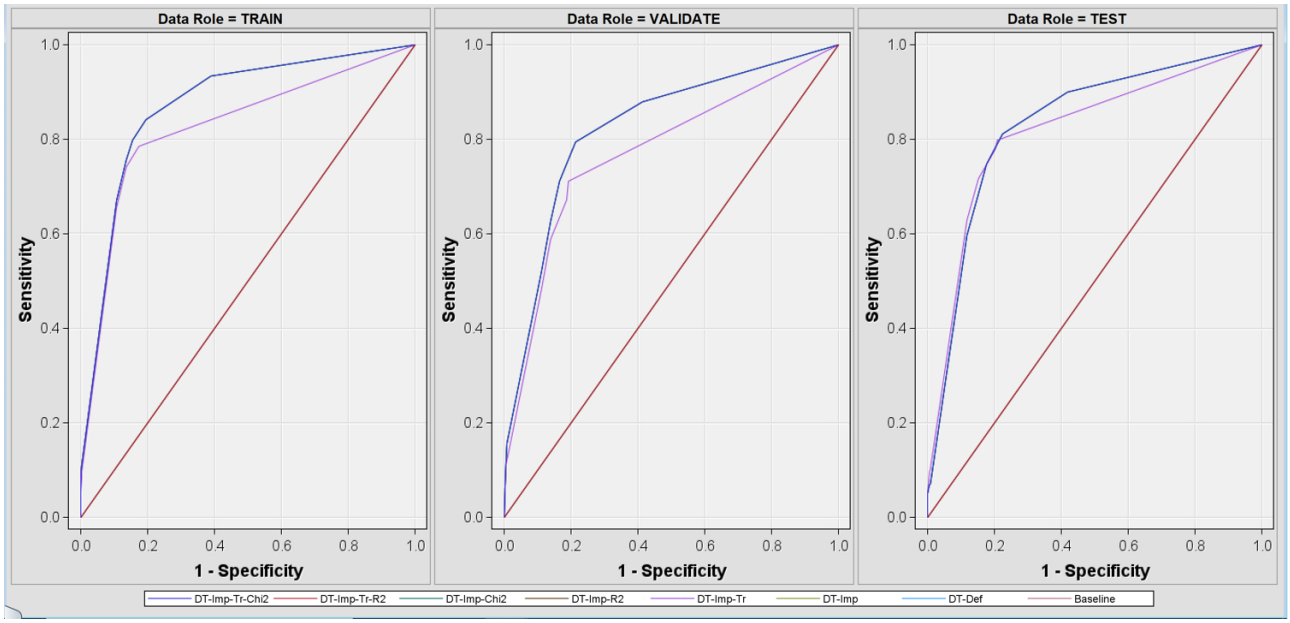


Figure 16. Roc Chart for Decission Tree Model

Fit Statistics
Model Selection based on Train: Roc Index (_AUR_)

Selected Model	Model Node	Model Description	Train: Roc Index	Train: Average Squared Error	Train: Misclassification Rate	Valid: Average Squared Error	Valid: Misclassification Rate
Y	Tree	DT-Def	0.870	0.07329	0.10564	0.08181	0.10487
	Tree2	DT-Imp	0.870	0.07329	0.10564	0.08181	0.10487
	Tree5	DT-Imp-Chi2	0.870	0.07329	0.10564	0.08181	0.10487
	Tree7	DT-Imp-Tr-Chi2	0.870	0.07329	0.10564	0.08181	0.10487
	Tree3	DT-Imp-Tr	0.824	0.07613	0.10785	0.08675	0.10783
	Tree4	DT-Imp-R2	0.500	0.10181	0.11504	0.10137	0.11448
	Tree6	DT-Imp-Tr-R2	0.500	0.10181	0.11504	0.10137	0.11448

Figure 17. Model selection statistics

Critical path of best model

The critical path, comprising nodes 1, 2, 5, and 10, is pivotal due to its lower validation count, signifying substantial influence on model predictions. Its identification underscores key decision points crucial for predictive accuracy, warranting attention for refined model refinement and enhanced performance in the report.

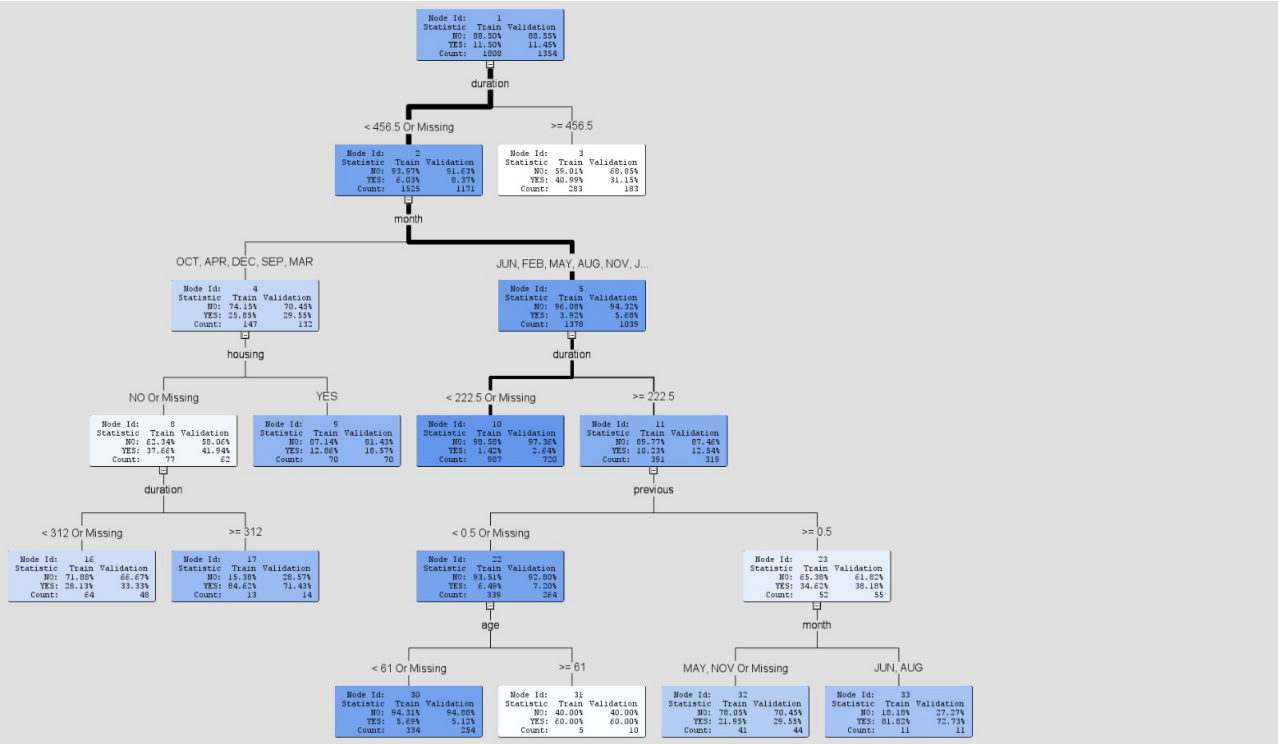


Figure 18. Critical Path

Target path of interest

The target path of interest resides in Node 10, with a validation percentage of 97.36% for non-subscribers and 2.64% for subscribers, highlighting its significance in predicting non-subscription. This path offers valuable insights for refining marketing strategies to target potential term deposit subscribers effectively.

Overfitting analysis

The analysis illustrates a common trend: as decision tree complexity increases, train misclassification decreases, while validation misclassification initially drops but later rises, indicating potential overfitting. Optimal complexity occurs with around 9 leaves before overfitting becomes prominent.

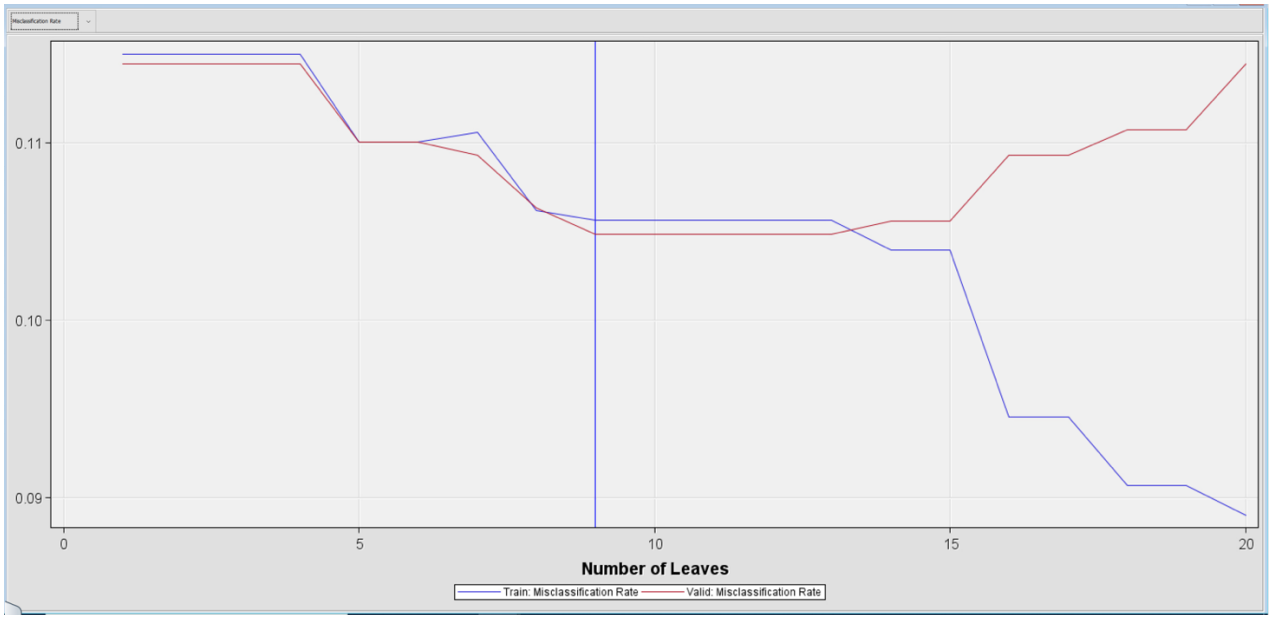


Figure 19. Overfitting Analysis (Misclassification Rate)

Analysis of the best model

Table 3 Summary results of the best performing models

Model	ROC Index	Cu. Lift	Scope %	True – %	False – %	True + %	False + %
Reg-IMP-Default	0.884	3.191192	**20**	**98.4987**	86.4516	**13.5484**	1.5013
DT_Def	0.87	3.174342	**20**	**99.0826**	84.5161	**15.4839**	0.9174

The analysis highlights "Reg-IMP-Default" as the superior model, boasting a higher ROC Index (0.884) and cumulative lift (3.191192) compared to "DT_Def" (0.87 and 3.174342, respectively). "Reg-IMP-Default" also demonstrates a higher true positive rate (86.4516%) and a lower false negative rate (13.5484%), making it the preferred choice for optimizing term deposit subscription predictions.

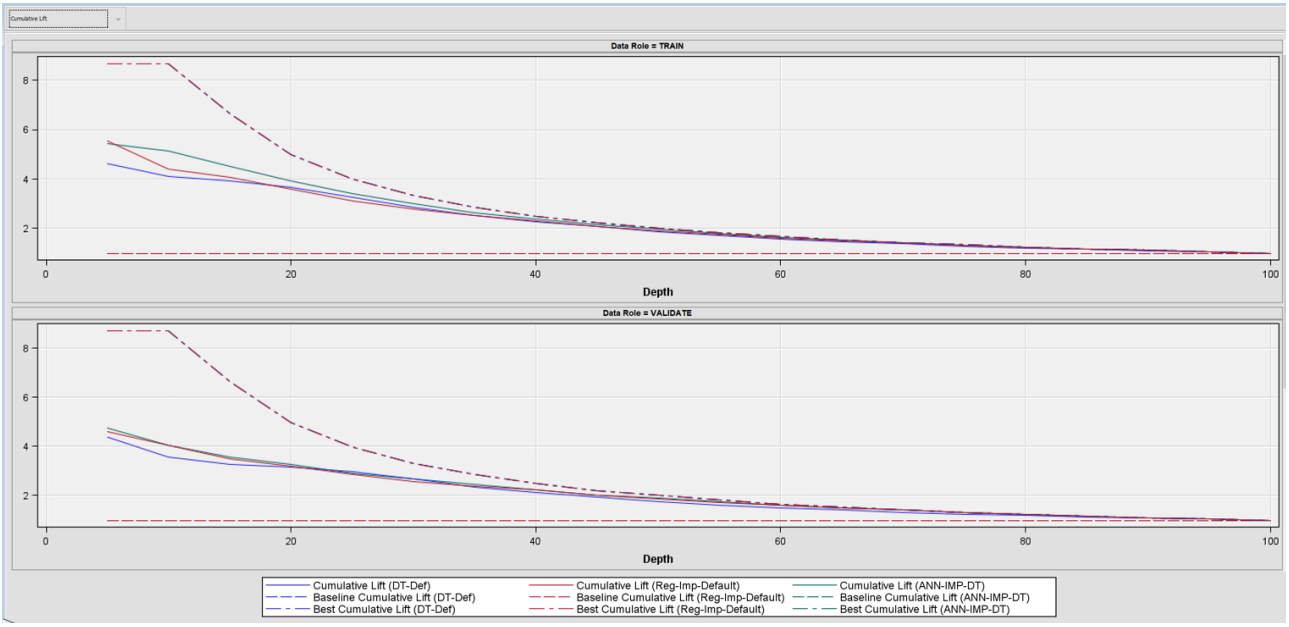


Figure 20. Score Ranking Overlay: y (Cumulative lift) for the best models

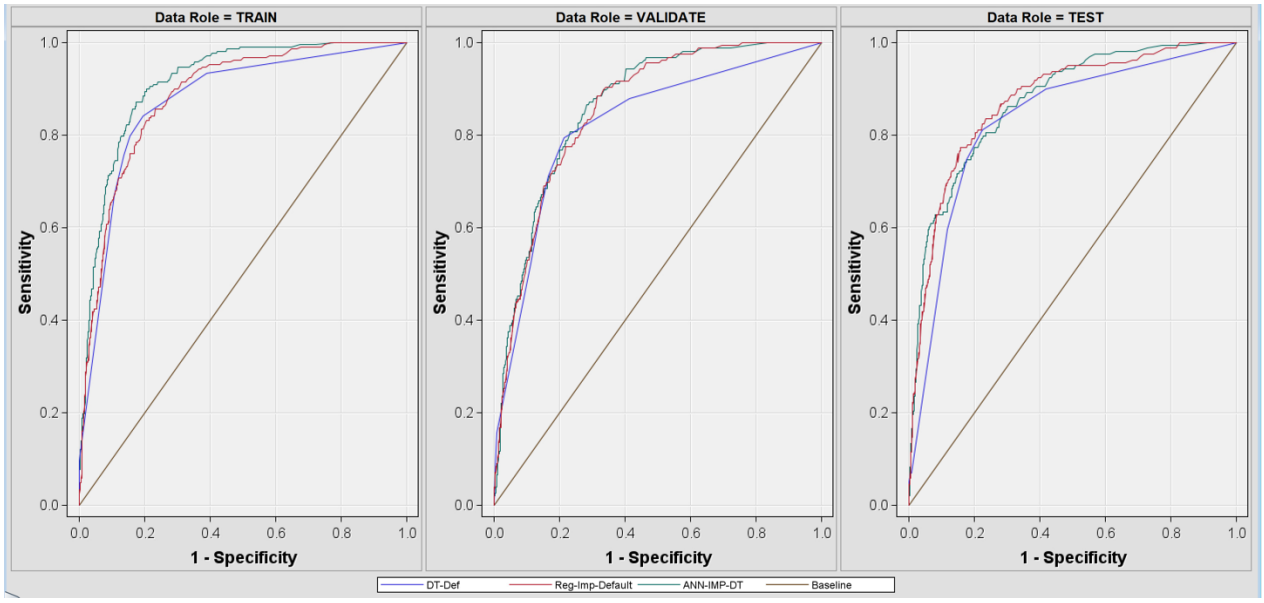


Figure 21. Roc Chart for the best models

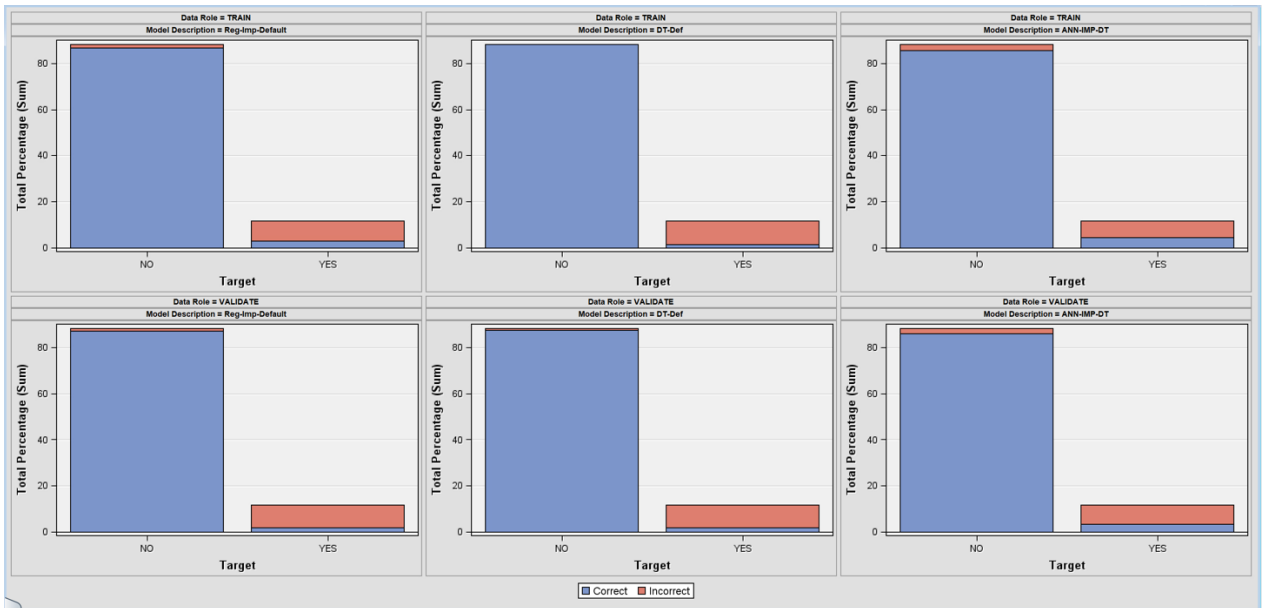


Figure 22. Classification chart where regression is best model.

Discussion on the breadth of areas of application and research in data mining

Several new trends are changing the landscape in the quickly developing field of predictive modeling. The blending of domain-specific knowledge with machine learning techniques is one such approach. The goal of this multidisciplinary strategy is to improve interpretability and predictive accuracy, especially in industries like marketing, finance, and healthcare. This combination of approaches offers potential paths forward for improving predictive modeling capabilities by utilizing domain expertise in feature selection, model construction, and outcome interpretation.

Deep learning architectures and algorithms are being developed, which is another significant trend. Because they can handle complicated, high-dimensional data types including text, pictures, and time-series data, techniques like convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are becoming more and more popular. Continuous investigations aim to enhance these deep learning models even more, with potential uses in financial forecasting, picture identification, natural language processing, and beyond.

An increasing focus is on combating bias, ensuring equity, and enhancing transparency in anticipatory modeling, notably in sensitive areas like law enforcement and loan approval. Efforts aim to reduce prejudice and maintain fairness in algorithmic decision-making, while endeavors strive to improve the transparency and comprehensibility of anticipatory models. Understandable AI techniques clarify model forecasts, bolstering confidence among stakeholders. Furthermore, the advent of large and continuous data technologies drives research into scalable predictive modeling methods to manage expanding datasets and dynamic data streams.

Conclusion

Significant data for the Portuguese bank's direct marketing campaigns to boost term deposit memberships was obtained from the BANK dataset analysis conducted with SAS Enterprise Miner. Models that used feature selection and imputation methods, such as Chi-square testing, performed well. The best predictor turned out to be the regression model with imputation using default settings. Nevertheless, there were differences in decision tree models' efficiency. New developments in deep learning architectures and the incorporation of domain-specific knowledge into predictive modeling offer potential directions for further study that will help the bank improve its marketing tactics and increase consumer interaction.

Recommendation

The regression model with default imputation proved most effective, suggesting prioritization in future campaigns. Decision tree model variations require further investigation for performance disparities. Embracing advancements in predictive modeling, especially deep learning, alongside domain-specific knowledge, can enhance interpretability and predictive accuracy, offering opportunities for optimization.

My Reflections on the process – What did I learn from this exercise?

Exploring forecasting modeling and data analysis using SAS Enterprise Miner and the BANK dataset deepened my understanding of SEMMA. I honed skills in feature selection, data imputation, and model evaluation, bolstering my expertise in predictive modeling. This exercise provided valuable insights into data-centric marketing strategies, enriching my analytical capabilities.

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Appendix

Variable name	Variable description
age	Client's age
job	Type of job ('admin.','blue-collar', 'entrepreneur', 'housemaid', 'management', 'retired', 'self-employed', 'services', 'student', 'technician', 'unemployed', 'unknown')
marital	Marital status ('divorced', 'married', 'single', 'unknown'; note: 'divorced' means divorced or widowed)
education	Education levels in Portugal ('basic.4y', 'basic.6y', 'basic.9y', 'high.school', 'illiterate', 'professional.course', 'university.degree', 'unknown')
default	has credit in default?
balance	average yearly balance
housing	has housing loan?
loan	has personal loan?
contact	contact communication type ('cellular', 'telephone')
day_of_week	last contact day of the week
month	last contact month of year ('jan', 'feb', 'mar', ..., 'nov', 'dec')
duration	last contact duration in seconds.
campaign	number of contacts performed during this campaign for this client (includes last contact)
pdays	number of days that passed by after the client was last contacted from a previous campaign (-1 means client was not previously contacted)
previous	number of contacts performed **before this campaign** for this client
poutcome	outcome of the **previous marketing campaign** ('failure', 'nonexistent', 'success')
y	has the client subscribed a term deposit?

Figure 23. Structure of Bank Data Set

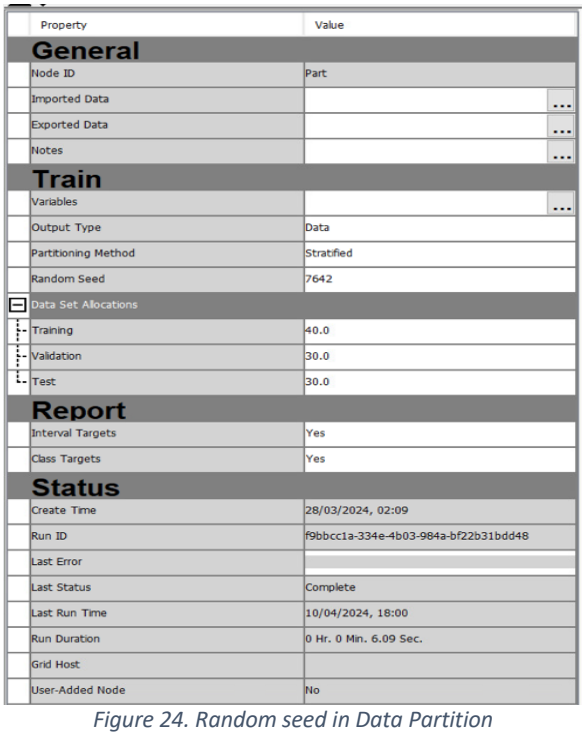


Figure 24. Random seed in Data Partition

Data Mining Roles

Data exploration, modification, modeling, and assessment are some of the tasks involved in the campaign optimization data mining process. Modification guarantees data quality, modeling builds predictive models, evaluation assesses model performance, and exploration finds important factors. Collectively, these positions maximize the efficacy of marketing campaigns by utilizing data insights.

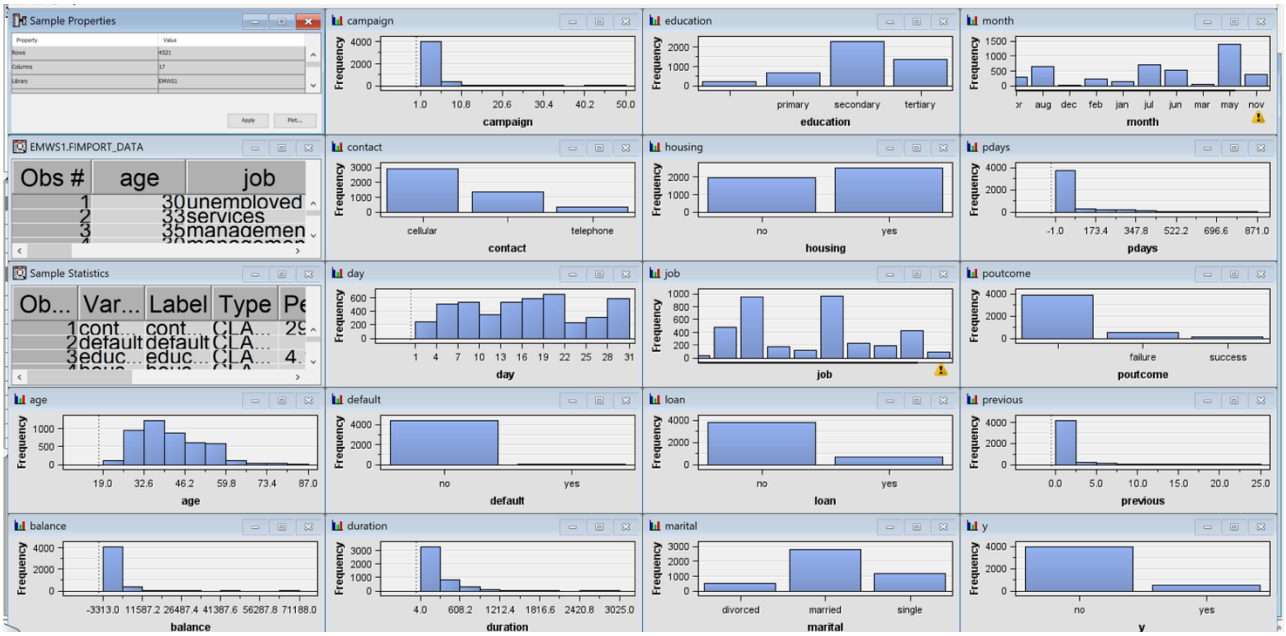


Figure 4. Histogram of all input and target variables

Observation Number	age	marital	default	housing	loan	month	y	Imputed: contact	Warnings	Imputed: education	Imputed: job	Tr_pdays	Tr_duration	Tr_balance	Tr_previous	Tr_day	Tr_campaign
1	30	married	no	yes	yes	jun	no	cellular		tertiary	management		2.298853	3.16938	0	1.732051	0.60206
2	35	single	no	no	no	feb	no	cellular		tertiary	management	5.17615	2.149219	2.873902	0.60206	4.795832	0.30103
3	39	married	no	yes	no	may	no	cellular		secondary	technician		2.179977	2.170262	0	2.44949	0.30103
4	39	married	no	yes	no	may	no	cellular		secondary	services		2.436163	3.971971	0	4.472136	0
5	40	married	no	no	yes	aug	no	cellular		tertiary	management		2.276462	2.290035	0	5.385165	0.30103
6	56	married	no	no	no	aug	no	cellular		secondary	technician		2.378398	3.610021	0	5.196152	0.69897
7	42	divorced	no	no	no	nov	no	cellular		tertiary	management		2.146128	1.230449	0	4.358899	0.477121
8	41	married	no	no	no	nov	no	cellular		tertiary	management		2.260071	3.769673	0	4.472136	0.30103
9	55	married	no	yes	no	may	no	cellular		primary	blue-collar		2.392697	2.79796	0	2.236068	0
10	56	married	no	no	yes	jul	no	cellular		secondary	self-employed		2.173186	2.89487	0	5.477226	0.30103
11	53	married	no	no	yes	aug	no	cellular		secondary	admin.		1.869232	2.025306	0	4.582576	0.30103
12	68	divorced	no	no	no	jul	yes	telephone		secondary	retired		2.952792	3.622214	0	3.741657	0.30103
13	59	married	no	no	no	nov	no	cellular		secondary	management		1.60206	1.653468	0	4.582576	0
14	32	single	no	yes	no	aug	yes	cellular		tertiary	management		2.981366	3.40432	0	5.09902	0.778151
15	49	married	no	no	no	aug	yes	cellular		tertiary	technician		2.549003	3.092018	0	3.605551	0.477121
16	42	divorced	no	yes	no	may	no	cellular		secondary	admin.		2.176091	3.258158	0	3.741657	0
17	78	divorced	no	no	no	oct	yes	telephone		primary	retired		1.986772	2.361728	0	4.690416	0
18	32	married	no	yes	no	nov	yes	cellular		secondary	blue-collar		2.129574	3.320146	0	3.741657	0
19	52	married	no	yes	no	may	no	cellular		secondary	blue-collar		2.562293	3.048442	0	3.605551	0
20	32	single	no	yes	no	nov	no	cellular		tertiary	technician		1.041393	3.343409	0	4.582576	0.60206
21	26	married	no	yes	no	may	no	cellular		primary	blue-collar		2.628389	0	0	4.582576	0
22	37	single	no	yes	no	aug	no	cellular		secondary	technician		3.240549	2.359835	0	4.472136	0.30103
23	36	single	no	yes	no	may	no	cellular		secondary	retired		2.49693	2.0086	0	5.09902	0
24	54	divorced	no	yes	yes	may	no	cellular		secondary	technician		2.762679	2.89487	0	3.872983	0
25	41	married	no	no	yes	jul	no	telephone		primary	blue-collar		2.74351		0	2.828427	0.477121
26	56	married	no	no	no	oct	no	cellular		secondary	admin.		2.348305	4.227218	0	2.645751	0
27	51	divorced	no	yes	no	may	no	cellular		secondary	blue-collar	5.141664	2.127105	2.30963	0.778151	2.828427	0
28	41	married	no	no	yes	jul	no	cellular		secondary	management		2.799341	0	0	2.645751	0.477121
29	31	single	no	yes	no	may	no	cellular		tertiary	management		2.187521	3.795811	0	3	0.30103
30	41	married	no	yes	no	jun	no	cellular		tertiary	services		2.187521	2.61595	0	1.732051	0.69897
31	40	married	no	yes	no	nov	no	cellular		secondary	unemployed	5.283204	2.30963	2.342423	0.30103	4.123106	0.30103
32	27	divorced	no	yes	no	jul	yes	cellular		secondary	admin.		2.814248	2.655138	0	4	0
33	37	married	no	no	no	jun	yes	cellular		tertiary	management		2.350248	3.119256	0	5.477226	0
34	34	married	no	yes	no	may	no	cellular		secondary	blue-collar		2.307496	3.262925	0	4.472136	0.30103
35	57	married	no	no	no	aug	no	cellular		secondary	blue-collar		2.363612	4.149034	0	3.316625	0.69897
36	41	married	no	yes	no	may	no	cellular		tertiary	admin.		2.173186	3.73456	0	3.741657	0.30103
37	30	single	no	no	no	oct	no	cellular		tertiary	admin.	4.927254	2.367356	2.418301	1.322219	4.358899	0
38	36	married	no	yes	no	may	no	cellular		secondary	blue-collar		2.52763	0	0	2.44949	0
39	49	married	no	no	no	nov	no	telephone		secondary	technician		2.537819	3.777934	0	4.582576	0.30103
40	57	single	no	yes	no	may	no	telephone		primary	services		1.812913	3.577262	0	3.605551	0.30103
41	32	single	no	yes	no	apr	no	cellular		tertiary	management	4.983607	2.4133	2.759668	0.60206	3.741657	0.30103
42	33	married	no	yes	no	feb	no	cellular		tertiary	self-employed		2.447158	2.684845	0	1.732051	0.30103
43	27	married	no	yes	no	jul	no	cellular		primary	housemaid		2.638489	0	0	4.795832	0.477121
44	21	single	no	no	no	jun	yes	cellular		secondary	student	5.135798	2.41162	3.396025	0.60206	5.477226	0.778151
45	33	single	no	no	no	jan	no	cellular		secondary	blue-collar	5.337538	1.880914	1.361728	0.30103	5.477226	0.30103
46	25	single	no	no	yes	nov	yes	cellular		secondary	technician		2.586587	2.704151	0	4.123106	0.30103
47	22	single	no	no	yes	aug	no	cellular		secondary	admin.		1.812913	3.614053	0	4.358899	0
48	39	married	no	no	no	aug	no	cellular		primary	blue-collar		1.838849	1.724276	0	3.605551	0
49	46	divorced	no	no	no	jun	no	cellular		tertiary	management		2.751279	1.414973	0	4.123106	0.30103
50	31	divorced	no	yes	no	jul	no	cellular		secondary	admin.		2.759377	3.276692	0	4.582576	0
51	45	married	no	no	no	aug	no	cellular		primary	blue-collar		2.053078	2.623249	0	4.242641	0.60206
52	32	married	no	yes	no	may	no	cellular		secondary	technician		1.20412	3.430398	0	4.582576	0
53	32	single	no	no	no	feb	no	telephone		secondary	blue-collar	5.666427	2.245513	2.359835	0.60206	2.236068	0
54	77	divorced	no	no	no	apr	yes	cellular		tertiary	retired		2.206826	3.668386	0	3.741657	0
55	41	married	no	yes	no	nov	no	cellular		primary	blue-collar		2.271842	2.227887	0	4.242641	0.477121
56	43	married	yes	yes	no	may	no	cellular		primary	blue-collar		1.826075		0	5.477226	0.477121

Figure 26. Sample Data

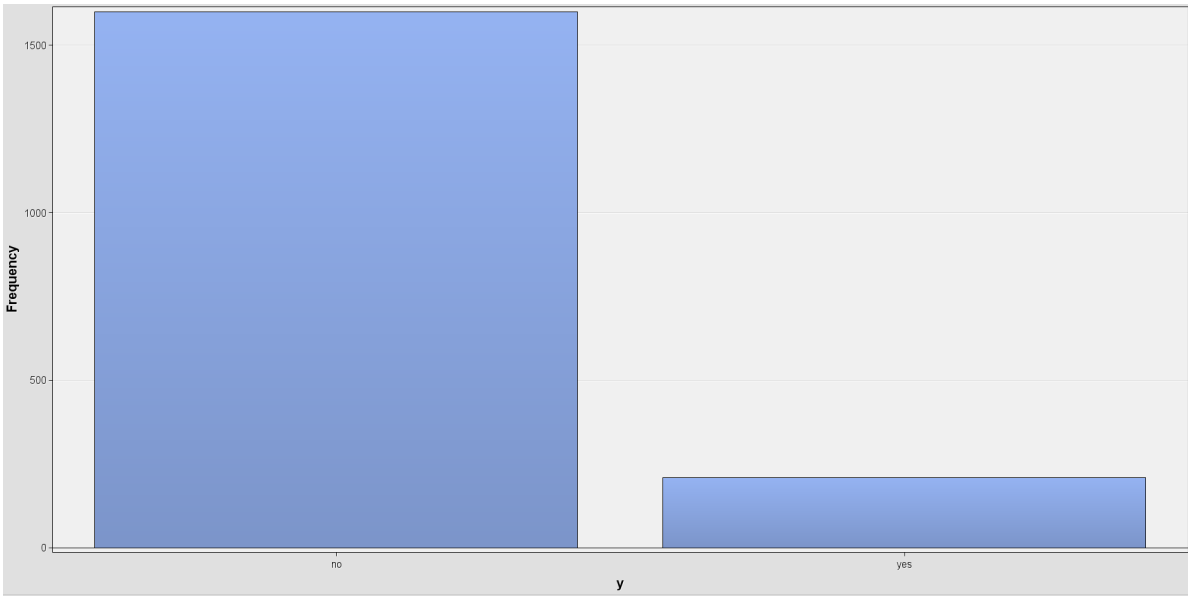


Figure 27. Frequency of target variable (After Sampling)

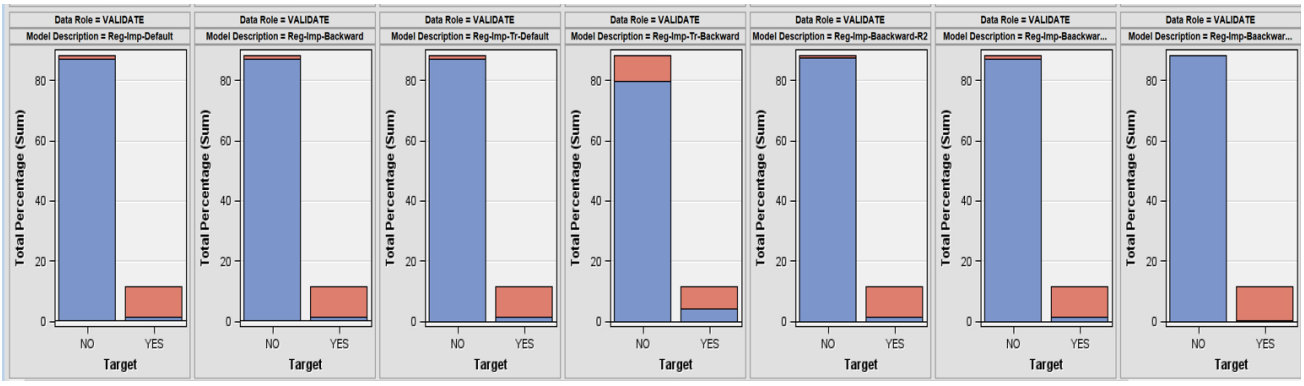


Figure 28. Classification Chart to select the best Regression.

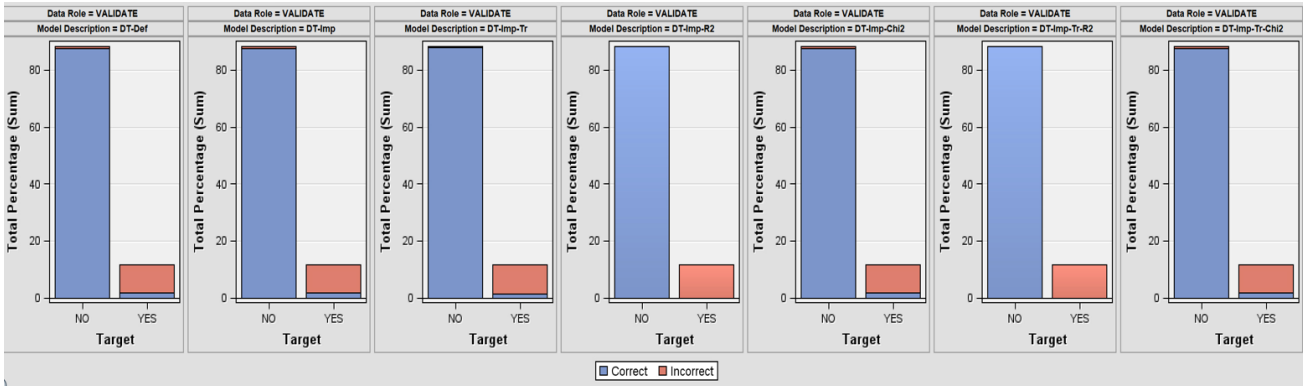
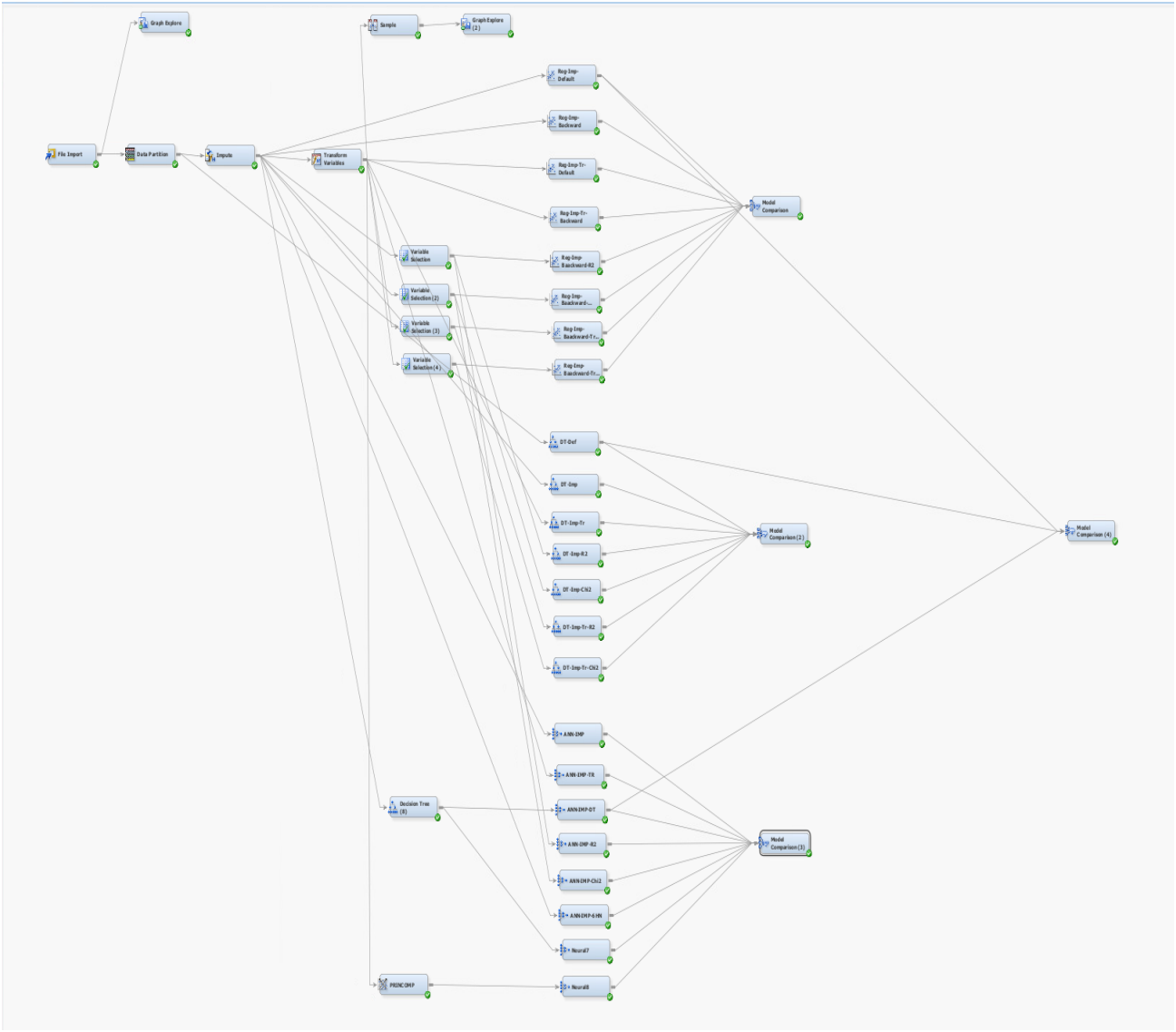


Figure 29. Classification Chart to select the best Neural Network

Workflow diagram


Graph Explore
Sample
Graph Explore (2)
File Import
Data Partition
Impute
Transform Variables
Reg-Imp-Default
Reg-Imp-Backward
Reg-Imp-Tr-Default
Reg-Imp-Tr-Backward
Model Comparison
Variable Selection
Variable Selection (2)
Variable Selection (3)
Variable Selection (4)
Reg-Imp-Backward-R2
Reg-Imp-Backward-...
Reg-Imp-Backward-Tr...
Reg-Imp-Backward-Tr...
DT-Def
DT-Imp
DT-Imp-Tr
DT-Imp-R2
DT-Imp-Chi2
DT-Imp-Tr-R2
DT-Imp-Tr-Chi2
Model Comparison (2)
Model Comparison (4)
ANN-IMP
ANN-IMP-TR
ANN-IMP-DT
ANN-IMP-R2
ANN-IMP-Chi2
ANN-IMP-HN
Neural7
Neural8
Decision Tree (8)
PRINCOMP
Model Comparison (3)